SVKM's NMIMS
Mukesh Patel School of Technology Management & Engineering
Computer Engineering Department
Program: MCA, Semester I

Course: Java Programming

Experiment No.01

PART A

(PART A : TO BE REFFERED BY STUDENTS)

A.1 Aim: To understand and implement basic concepts like Data types, Variables and Operators in Java.

TASK 1: Write a Java program to take input for two integer numbers from the user and add them.

Task 2: The straight-line method of computing the yearly logical depreciation of the value of an item is given by

Depreciation = (purchase-price –Salvage Value)/years of service

Write a program to determine the salvage value of an item when the purchase price, years of service, and the annual depreciation are given.

Task 3: The total distance travelled by a vehicle in t seconds is given by

Distance = $ut + (at^2)/2$

Where u is the initial velocity (meters per second), a is the acceleration (meters per second^2). Write a program to evaluate the distance travelled at regular intervals of time, given the values of u and a.

Task 4: Write a Java program to reverse the number entered by the user.

TASK 5: Consider the scenario of processing marks of a student John for a course as part of a student management system. The assumption is that John takes exams in five different subjects. Find the total marks and average marks scored by John.

Task 6: Write a Java program to perform basic calculator operations.

Task 7: Write a Java program to swap two numbers.

Task 8: Write a Java program to check if entered number is odd or even.

Task 9: Write a Java program to find largest of three numbers entered by the user.

Task 10: Write a Java program to check whether a number is positive, negative or zero.

Task 11: Write a Java program to check if two of three Boolean variables are true.

Task 12: Write a Java program to check whether an alphabet is vowel or consonant.

Task 13: Write a Java program whether entered year is leap year or not.

Task 14: Write a Java program that calculates and displays the amount of money you would have if you invested 10000 Rs at 5% interest for one year. Use the formula:

Future Amount = Principal*Rate*Time.

A.2 Prerequisite:

1. Fundamental concepts of C\C++.

A.3 Outcome:

After successful completion of this experiment, students will be able to

1. Implement concept of Data types, variables in Java

A.4 Theory:

A.4.1.

Java Variables

A variable is a container which holds the value while the Java program is executed. A variable is assigned with a data type.

Variable is a name of memory location. There are three types of variables in java: local, instance and static. There are two types of data types in Java: primitive and non-primitive.

Variable

A variable is the name of a reserved area allocated in memory. In other words, it is a name of the memory location. It is a combination of "vary + able" which means its value can be changed.

```
int data=50;//Here data is variable
```

There are three types of variables in Java

- local variable
- instance variable
- static variable

Data Types in Java

Data types specify the different sizes and values that can be stored in the variable. There are two types of data types in Java:

1. **Primitive data types:** The primitive data types include boolean, char, byte, short, int, long, float and double.
2. **Non-primitive data types:** The non-primitive data types include Classes, Interfaces, and Arrays.

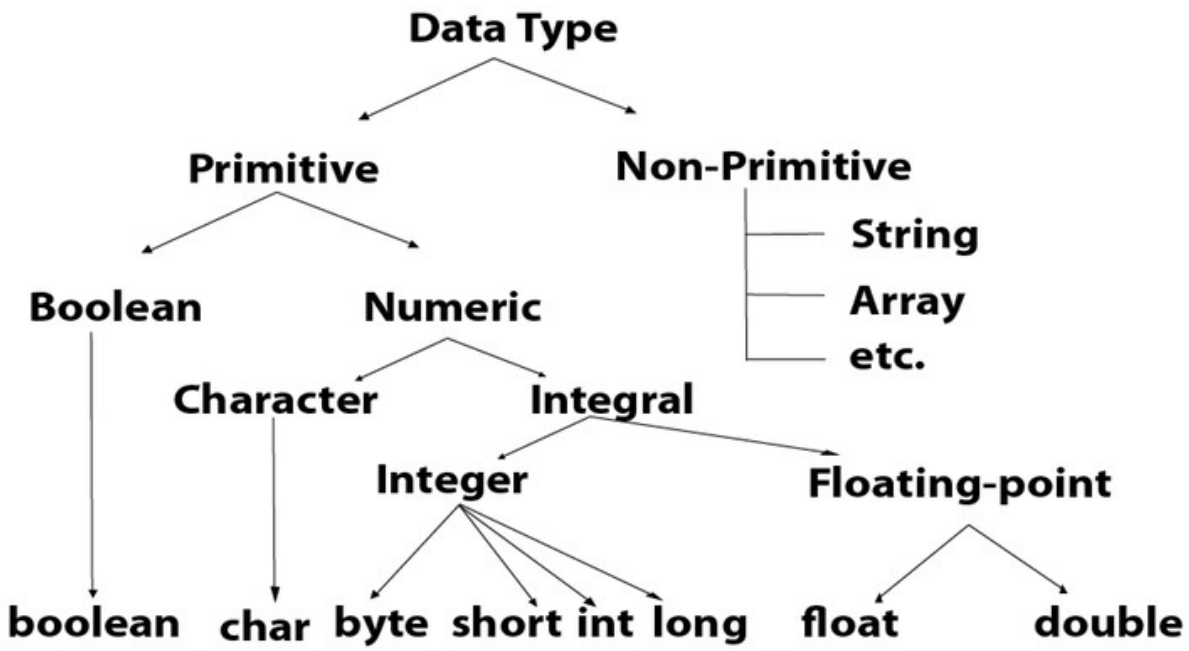
Java Primitive Data Types

In Java language, primitive data types are the building blocks of data manipulation. These are the most basic data types available in Java language.

Java is a statically-typed programming language. It means, all variables must be declared before its use. That is why we need to declare variable's type and name.

There are 8 types of primitive data types:

- boolean data type
- byte data type
- char data type
- short data type
- int data type
- long data type
- float data type
- double data type


Data Type
Primitive
Non-Primitive
String
Array
etc.
Boolean
Numeric
Character
Integral
Integer
Floating-point
boolean
char
byte
short
int
long
float
double

PART B

(PART B : TO BE COMPLETED BY STUDENTS)

Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the portal at the end of the practical. The filename should be **MCA_JP_rollno_experimentno Example : MCA_JP_C001_Exp1**

Roll No. A012	Name: Brinda Das
Class : MCA	Batch : B1
Date of Experiment: 21-07-2026	Date of Submission: 21-07-2026
Grade :	

B.1 Software Code written by student:

Task 1: Write a Java program to take input for two integer numbers from the user and add them.

add.java

```
import java.util.*;
public class add {
    public static void main(String[] args) {
        int i,j,res;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter first number: ");
        i=sc.nextInt();
        System.out.println("Enter second number: ");
        j=sc.nextInt();
        res=i+j;
        System.out.println("The addition of "+i+" and "+j+" is: "+res);
    }
}
```

```
C:\Users\mpstme.student\Desktop\JAVA A012>javac add.java

C:\Users\mpstme.student\Desktop\JAVA A012>java add
Enter first number:
7
Enter second number:
15
The addition of 7 and 15 is: 22
```

Task 2: Write a program to determine the salvage value of an item when the purchase price, years of service, and the annual depreciation are given.

Depreciation.java

```
import java.util.Scanner;
public class Depreciation {
    public static void main(String[] args) {
        int Depreciation_value, salvage_value, purchase_price, years_of_service;
        Scanner sc =new Scanner(System.in);
        System.out.println("Enter the Depreciation value: ");
        Depreciation_value=sc.nextInt();

        System.out.println("Enter the purchase price: ");
        purchase_price=sc.nextInt();

        System.out.println("Enter the years of service: ");
        years_of_service=sc.nextInt();

        salvage_value=purchase_price-(Depreciation_value*years_of_service);

        System.out.println("The salvage value of the particular purchase item is:
"+salvage_value);
        }
}
```

```
C:\Users\mpstme.student\Desktop\JAVA A012>javac Depreciation.java

C:\Users\mpstme.student\Desktop\JAVA A012>java Depreciation
Enter the Depreciation value:
900
Enter the purchase price:
12000
Enter the years of service:
3
The salvage value of the particular purchase item is: 9300
```

Task 3: Write a program to evaluate the distance travelled at regular intervals of time, given the values of u and a.

DistanceCalculator.java

```
import java.util.Scanner;

public class DistanceCalculator {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter initial velocity (u) in m/s: ");

        double u = scanner.nextDouble();

        System.out.print("Enter acceleration (a) in m/s^2: ");

        double a = scanner.nextDouble();

        System.out.print("Enter total time (in seconds): ");

        int total Time = scanner.nextInt();

        System.out.print("Enter time interval (e.g., 1, 2, 5 seconds): ");

        int interval = scanner.nextInt();

        System.out.println("\n--- Distance Results ---");

        for (int t = 0; t <= totalTime; t += interval) {

            double distance = (u * t) + (a * t * t) / 2.0;

       System.out.println("At time t = " + t + " seconds, Distance = " +
      distance + " meters");

            }

        scanner.close();

    }

}
```

```
D:\Java>javac DistanceCalculator.java

D:\Java>java DistanceCalculator
Enter initial velocity (u) in m/s: 0
Enter acceleration (a) in m/s^2: 3
Enter total time (in seconds): 10
Enter time interval (e.g., 1, 2, 5 seconds): 3

--- Distance Results ---
At time t = 0 seconds, Distance = 0.0 meters
At time t = 3 seconds, Distance = 13.5 meters
At time t = 6 seconds, Distance = 54.0 meters
At time t = 9 seconds, Distance = 121.5 meters
```

Task 4: Write a Java program to reverse the number entered by the user.

reverse.java

```
import java.util.Scanner;

public class reverse {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int number = scanner.nextInt();

        int reversed = 0;

        while (number != 0) {

            int lastDigit = number % 10;

            reversed = (reversed * 10) + lastDigit;

            number = number / 10;

        }

        System.out.println("Reversed number: " + reversed);

        scanner.close();

    }

}
```

```
D:\Java>javac reverse.java

D:\Java>java reverse
Enter a number: 123
Reversed number: 321
```

Task 5: Consider the scenario of processing marks of a student John for a course as part of a student management system. The assumption is that John takes exams in five different subjects. Find the total marks and average marks scored by John.

StudentMarks.java

```
import java.util.Scanner;

public class StudentMarks {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter John's marks for 5 subjects:");

        double mark1 = scanner.nextDouble();

        double mark2 = scanner.nextDouble();

        double mark3 = scanner.nextDouble();

        double mark4 = scanner.nextDouble();

        double mark5 = scanner.nextDouble();

        double total = mark1 + mark2 + mark3 + mark4 + mark5;

        double average = total / 5;

        System.out.println("John's Result: ");

        System.out.println("Total Marks: " + total);

        System.out.println("Average Marks: " + average);

        scanner.close();

    }

}
```

```
D:\Java>javac StudentMarks.java

D:\Java>java StudentMarks
Enter John's marks for 5 subjects:
45
55
44
40
35
John's Result:
Total Marks: 219.0
Average Marks: 43.8
```

Task 6: Write a Java program to perform basic calculator operations.

calculator.java

```
import java.util.Scanner;

public class calculator {
    public static void main(String[] args) {
        int a, b, op, res;
        Scanner sc=new Scanner(System.in);
        System.out.println("Calculator \n Enter first number: ");
        a=sc.nextInt();

        System.out.println("Enter second number: ");
        b=sc.nextInt();

        System.out.println("Select operator \n1. Add \n2. Subtract \n3. Divide
\n4. Multiply \n5.Modolus");
        op=sc.nextInt();

        switch(op){
            case 1:
                res=a+b;
                System.out.println("Addition of "+a+" and "+b+" is "+res);
                break;
            case 2:
                res=a-b;
                System.out.println("Subtraction of "+a+" and "+b+" is "+res);
                break;
            case 3:
                res=a/b;
                System.out.println("Division of "+a+" and "+b+" is "+res);
```

```
                break;
            case 4:
                res=a*b;
                System.out.println("Multiplication of "+a+" and "+b+" is "+res);
                break;
            case 5:
                res=a%b;
                System.out.println("Modolus of "+a+" and "+b+" is "+res);
                break;
            default:
                System.out.println("Invalid operator");
        }
    }
}
```

```
C:\Users\mpstme.student\Desktop\JAVA A012>javac calculator.java

C:\Users\mpstme.student\Desktop\JAVA A012>java calculator
Calculator
 Enter first number:
15
Enter second number:
7
Select operator
1. Add
2. Subtract
3. Divide
4. Multiply
5.Modolus
2
Subtraction of 15 and 7 is 8
```

Task 7: Write a Java program to swap two numbers.

numswap.java

```
import java.util.Scanner;

public class numswap {
    public static void main(String[] args) {
        int a, b, temp;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter first number: ");
        a=sc.nextInt();
        System.out.println("Enter second number: ");
        b=sc.nextInt();
```

```
        System.out.println("Before swapping the values are: a ="+a+" and b =
"+b);
       temp = a;
        a = b;
        b = temp;
        System.out.println("Before swapping the values are: a ="+a+" and b =
"+b);
    }
}
```

```
C:\Users\mpstme.student\Desktop\JAVA A012>javac numswap.java

C:\Users\mpstme.student\Desktop\JAVA A012>java numswap
Enter first number:
11
Enter second number:
27
Before swapping the values are: a =11 and b = 27
Before swapping the values are: a =27 and b = 11
```

Task 8: Write a Java program to check if entered number is odd or even.

OddEven.java

```
import java.util.Scanner;
public class OddEven {
    public static void main(String[] args) {
        int i;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter a number: ");
        i=sc.nextInt();
        if(i%2 == 0){
            System.out.println("The number "+i+" is even");
        }
        else{
            System.out.println("The number "+i+" is odd");
        }
    }
}
```

```
C:\Users\mpstme.student\Desktop\JAVA A012>javac OddEven.java

C:\Users\mpstme.student\Desktop\JAVA A012>java OddEven
Enter a number:
7
The number 7 is odd

C:\Users\mpstme.student\Desktop\JAVA A012>java OddEven
Enter a number:
8
The number 8 is even
```

Task 9: Write a Java program to find largest of three numbers entered by the user.

Large.java

```
import java.util.Scanner;
public class Large {
    public static void main(String[] args) {
        int a, b, c;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter first number: ");
        a=sc.nextInt();
        System.out.println("Enter second number: ");
        b=sc.nextInt();
        System.out.println("Enter third number: ");
        c=sc.nextInt();
        if(a>=b && a>=c){
            System.out.println("Number "+a+" is larger than "+b+" and "+c);
        }
        else if(b>=a && b>=c)
        {
            System.out.println("Number "+b+" is larger than "+a+" and "+c);
        }
        else if(c>=a && c>=b){
            System.out.println("Number "+c+" is larger than "+b+" and "+a);
        }
        else
        {
            System.out.println("All nuumbers are equal: "+a+" "+b+" "+c);
        }
    }}
```

```
C:\Users\mpstme.student\Desktop\JAVA A012>javac Large.java

C:\Users\mpstme.student\Desktop\JAVA A012>java Large
Enter first number:
5
Enter second number:
7
Enter third number:
15
Number 15 is larger than 7 and 5
```

Task 10: Write a Java program to check whether a number is positive, negative or zero.

PositiveNegZero.java

```
import java.util.Scanner;
public class PositiveNegZero {
    public static void main(String[] args) {
         int i;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter a number: ");
        i=sc.nextInt();
        if(i==0){
            System.out.println("The number is zero: "+i);
        }
        else if(i>0){
            System.out.println("The number is positive: "+i);
        }
        else
        {
            System.out.println("The number is negative: "+i);
        }

    }

}
```

```
C:\Users\mpstme.student\Desktop\JAVA A012>javac PositiveNegZero.java

C:\Users\mpstme.student\Desktop\JAVA A012>java PositiveNegZero
Enter a number:
0
The number is zero: 0

C:\Users\mpstme.student\Desktop\JAVA A012>java PositiveNegZero
Enter a number:
-9
The number is negative: -9

C:\Users\mpstme.student\Desktop\JAVA A012>java PositiveNegZero
Enter a number:
15
The number is positive: 15
```

Task 11: Write a Java program to check if two of three Boolean variables are true.

bool.java

```
import java.util.Scanner;

public class bool {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter first boolean (true/false): ");

        boolean a = scanner.nextBoolean();


        System.out.print("Enter second boolean (true/false): ");

        boolean b = scanner.nextBoolean();


        System.out.print("Enter third boolean (true/false): ");

        boolean c = scanner.nextBoolean();


        boolean result = (a && b) || (b && c) || (a && c);
```

```
        System.out.println("Are at least two true? " + result);
        scanner.close();
    }
}
```

```
D:\Java>javac bool.java

D:\Java>java bool
Enter first boolean (true/false): false
Enter second boolean (true/false): true
Enter third boolean (true/false): true
Are at least two true? true
```

Task 12: Write a Java program to check whether an alphabet is vowel or consonant.

vowel.java

```
import java.util.Scanner;
public class vowel
 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter an alphabet: ");
        char ch = scanner.next().toLowerCase().charAt(0);
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
            System.out.println(ch + " is a Vowel.");
        } else if ((ch >= 'a' && ch <= 'z')) {
            System.out.println(ch + " is a Consonant.");
        } else {
            System.out.println("Not a valid letter.");
        }
```

```
        scanner.close();
    }
}
```

```
D:\Java>javac vowel.java

D:\Java>java vowel
Enter an alphabet: b
b is a Consonant.

D:\Java>java vowel
Enter an alphabet: o
o is a Vowel.
```

Task 13: Write a Java program whether entered year is leap year or not.

leapyear.java

```
import java.util.Scanner;

public class leapyear {

public static void main(String[] args) {

Scanner scanner = new Scanner(System.in);

System.out.print("Enter a year: ");

int year = scanner.nextInt();

if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {

System.out.println(year + " is a Leap Year.");

} else {

System.out.println(year + " is NOT a Leap Year.");

}

scanner.close(); }}
```

```
D:\Java>javac leapyear.java

D:\Java>java leapyear
Enter a year: 2024
2024 is a Leap Year.

D:\Java>java leapyear
Enter a year: 2026
2026 is NOT a Leap Year.
```

Task 14: Write a Java program that calculates and displays the amount of money you would have if you invested 10000 Rs at 5% interest for one year.

investment.java

```
public class investment {

    public static void main(String[] args) {

        double principal = 10000;

        double rate = 5;

        int time = 1;

        double interest = (principal * rate * time) / 100;

        double futureAmount = principal + interest;

        System.out.println("Principal Amount: Rs. " + principal);

        System.out.println("Interest Earned: Rs. " + interest);

        System.out.println("Total Future Amount: Rs. " + futureAmount);

    }

}
```

```
D:\Java>javac investment.java

D:\Java>java investment
Principal Amount: Rs. 10000.0
Interest Earned: Rs. 500.0
Total Future Amount: Rs. 10500.0
```

B.2 Question of Curiosity:

1. Define Datatype, Variable and keywords.

- Variable: A named location in memory used to store data values during program execution.
    - Example: int x = 10; (x is the variable).
- Datatype: Defines the type and size of data a variable can hold.
    - Example: int stores 4 bytes of whole numbers.
- Keywords: Reserved words in Java that have special, predefined meanings to the compiler and cannot be used as variable names.
    - Example: class, public, static, void.

2. Explain different types of operators available in Java

Operators are symbols used to perform operations on variables and values.

1. Arithmetic Operators (+, -, *, /, %): Used to perform basic mathematical operations.
2. Relational Operators (==, !=, >, <, >=, <=): Compares two values and returns a Boolean output (true or false).
3. Logical Operators (&&, ||, !): Combines multiple conditions (AND, OR, NOT).
4. Assignment Operators (=, +=, -=): Used to assign or modify values in a variable.
5. Unary Operators (++, --): Increases or decreases a variable's value by 1.

3. Explain different types of data types available in Java.

Java data types are broadly classified into Primitive and Non-Primitive types:

A. Primitive Data Types (8 Built-in Types)

Stored directly in memory with fixed sizes.

- Numeric (Integers): byte, short, int, long
- Decimal (Floating point): float, double

- Character: char
- Boolean: boolean (true or false)

B. Non-Primitive (Reference) Data Types

Refers to memory addresses/objects created by programmers.

- String (text data)
- Arrays (collection of similar elements)
- Classes & Interfaces

4. Explain basic blocks of Java program.

Every standard Java program consists of five main components:

1. Package Declaration (Optional): Groups related classes together (e.g., package myCode;).
2. Import Statements: Imports built-in libraries (e.g., import java.util.Scanner;).
3. Class Header: Defines the class container (e.g., public class Main { ... }).
4. Main Method: Entry point where execution begins (public static void main(String[] args)).
5. Statements & Expressions: Executable instructions ending with a semicolon ;.

5. What is Java Virtual Machine?

A Java virtual machine (JVM) is a virtual machine that enables a computer to run Java programs as well as programs written in other languages, other languages referred to as JVM languages that are also compiled to Java bytecode. The JVM is detailed by a specification that formally describes what is required in a JVM implementation. Having a specification ensures interoperability of Java programs across different implementations so that program authors using the Java Development Kit (JDK) need not worry about idiosyncrasies of the underlying hardware platform.

B.3 Conclusion:

Through this assignment, I learned and practiced the basics of Java programming. By completing these 14 tasks and answering the core theory questions, I gained hands-on experience using variables, data types, and different operators. I built programs using decision-making statements (if-else) and loops, and learned how to take user input using the Scanner class to solve real-world math and logic problems. Overall, this assignment helped me understand how Java code is structured and executed, building a strong foundation for future coding tasks.

****